

A top-down view of a light teal desk. In the top left is a silver laptop with a black keyboard. To its right is a white mouse and a dark coffee cup. In the bottom left is a brown leather notebook with a white pen and a pair of tortoiseshell glasses.

AIClub





AI Project Cycle – Problem Scoping Stage

What is a Problem Statement?

- A Problem Statement is a clear and concise description of the issue that needs to be solved.

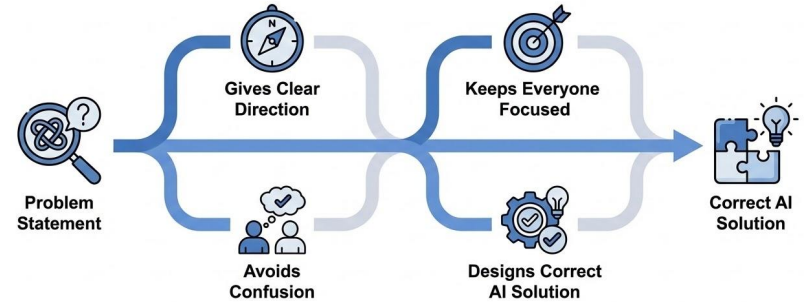
It explains:

- What the problem is
- Who is affected
- Why it needs to be solved



Why Do We Need a Problem Statement?

- A problem statement:
 - Gives clear direction
 - Avoids confusion
 - Keeps everyone focused
 - Helps design the correct AI solution



If the problem is unclear, the solution will also be unclear.

Characteristics of a Good Problem Statement

- A good problem statement should be:
 - Clear
 - Specific
 - Concise
 - Focused on the issue (not the solution)



It should NOT include the solution.

Template of a Problem Statement

- Basic Format:

Our [stakeholders] → **WHO**

has a problem that [issue] → **WHAT**

when / while [context] → **WHERE**

An ideal solution would [benefit] → **WHY**

Notice how it connects directly to 4Ws.



Example 1

- **Problem Statement:**

“The problem of late book returns affects students in the school library because it creates a shortage of available books.”

- **4Ws Identified:**

WHO → Library students

WHAT → Books are returned late

WHERE → School library

WHY → Causes shortage of books



Example 2

- **Problem Statement:**

“The problem of long waiting times affects patients at the hospital reception because it causes discomfort and delays in treatment.”

- **4Ws Identified:**

WHO → Patients

WHAT → Long waiting time

WHERE → Hospital reception

WHY → Causes discomfort and delays



Common Mistakes

- **Common mistakes include:**
 - Writing a solution instead of the problem
 - Being too vague
 - Missing stakeholders
 - Writing very long descriptions

Example (Incorrect):

“We should build an app to reduce waiting time.”

This is a solution, not a problem statement.



Classroom Activity

- Given 4Ws:
WHO | WHAT | WHERE | WHY ?
- Write 4Ws for all problems discussed in previous chapter:
 - Excessive mobile phone usage
 - Plastic waste in school
 - Low attendance



Classroom Activity

- **Answer:**
- 4Ws for Excessive mobile phone usage:
 - WHO: Students
 - WHAT: Students are using mobile phones too much, especially for games and social media.
 - WHERE: At home, in classrooms, and during study time
 - WHY: It affects students' focus, reduces study time, and can lower academic performance.

WHO?

WHAT?

WHERE?

WHY?

Classroom Activity

- **Answer:**
- 4Ws for Plastic Waste in School:
 - WHO: Students, teachers, and school staff
 - WHAT: Plastic waste such as bottles, wrappers, and bags is not disposed of properly.
 - WHERE: School campus, classrooms, playground, and cafeteria.
 - WHY: It causes pollution, makes the school environment dirty, and harms the environment.

WHO?

WHAT?

WHERE?

WHY?

Classroom Activity

- **Answer:**
- 4Ws for Problem 3: Low Attendance:
 - WHO: Students and teachers
 - WHAT: Many students are absent from school frequently.
 - WHERE: School classrooms.
 - WHY: Students miss important lessons, which affects their learning and academic performance.

WHO?

WHAT?

WHERE?

WHY?

Key Takeaways

- ✓ A problem statement clearly defines the issue to be solved.
- ✓ It is based on the 4Ws Canvas.
- ✓ It focuses on the problem, not the solution.
- ✓ A strong problem statement leads to a successful AI project.

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THANK YOU



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